

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 10

(1) Research Program/Project/Study Title: Utilization of Blue Ternate (Clitoria Ternatea) Concentrate	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
		Municipality/City:	Roxas City
		Province:	Capiz
		Region:	IV
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 2 of 10

(7) Sector/Commodity	(8) Discipline/Field of Specialization
	Food Engineering

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

This study aims to determine the feasibility of Blue Ternate (*Clitoria ternatea*) in the development of a new juice drink concentrate. A control and four treatments were created with the following concentrations (T1) 950g of concentrate and 50g of blue ternate, (T2) 900g of concentrate and 100g, (T3) . measure 850g of concentrate and 150g, and (T4) 800g of concentrate and 200g and (T5) with no blue ternate as control. An acceptability test using hedonic scale was conducted to determine the most preferred formulation with and the results showed T4 as the most preferred. A T4 sample was tested after 1-week and another was stored and tested after 1 year of storage. For 1-week T4 sample, results show it has an eminence color, slightly sour flavor, moderate perceivable aroma, and thickened consistency Tests for pH and TSS for physico-chemical profiling was conducted and showed results of 5.13 and 54.33 °B respectively. It was also tested for APC and had a result of 0 cfu/ml which is within the FDA standard for frozen juice concentrate. The 1-year sample was also tested for sensory and microbial tests. Results show it has a vivid violet color and slightly sweet after taste for flavor while its aroma and consistency remained the same. Test for APC revealed a result of <1 cfu/ml. A preference test was conducted for the two MATs and panelists preferred the 1-week sample. Thus, Blue Ternate is feasible in the development of juice drink and is safe to consume after 1 year of storage.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Blue ternate, concentrate, juice drink

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Food security has been a longstanding issue in the Philippines and has become even more prominent in the year 2022 (Candelaria, 2021). Food waste is also evident, and the marginalized cannot afford food supplies in the market (Bacilig, 2022). Providing all consumers with adequate food has been a challenge for the past few years, and many efforts have been made to address this issue. These challenges require people to be more innovative and creative in order to survive.

Philippines being an archipelagic country has various plants and herbs, many of which are rich in vitamins and minerals that can be used to address health issues present in the country. Utilization of local plants and herbs in food can result to more affordable products with high nutritional content. These products may be used to help address the food security and health issue in the country.

One of the local plants in the country is Blue ternate (*Clitoria ternatea*). Traditionally it used as a natural colorant due to its high level of anthocyanins and other flavonoids (Fuzetti, C. et.al., 2022). It is also known to contain the antioxidant, proanthocyanidin, which increases blood flow to the optic nerves. This maintains healthy eyesight and helps prevent glaucoma and other ocular ailments. Blue ternate also has significant values of minerals, fiber, protein, and carbohydrates indicating it can be used in pharmaceutical products (Deshmukh and Jadhav, 2014).

Since blue ternate has a variety of health benefits, it is usually added to hot water to create teas and juice drinks as home remedies or simply as a refreshing afternoon drink. Research and food product development in the country utilizing Blue ternate is only minimal. Three of which are "PROCESS OF PRODUCING BLUE TERNATE (CLITORIA TERNATE) FLOWER EXTRACT AS COLORANT FOR KAMIAS (AVERRHOA BILIMBI) FRUIT EXTRACT DISHWASHING LIQUID", "FORMULATION OF A BLUE TERNATE LEMONADE" and "PROCESS OF PRODUCING BLUE TERNATE LEMONADE"

Thus, this study aims to utilize Blue ternate in formulating a safe and acceptable alternative juice drink concentrate. The petals of the plant was the main raw material in this study together with local ingredients used in creating a fruit juice concentrate.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 10

Objectives

The research focus is to determine if Blue Ternate (*Clitoria ternatea*) is feasible in making Blue Ternate Juice Drink.

Specifically:

1. Determine the most acceptable treatment (MAT) of Blue Ternate Juice Drink based on
 - a. flavor
 - b. color
 - c. aroma
 - e. consistency
2. Determine the sensorial characteristics of the MAT - Blue Ternate Juice Drink based on
 - a. flavor
 - b. color
 - c. aroma
 - d. consistency
3. Determine the Aerobic Plate Count of the MAT- Blue Ternate Juice Drink
4. Determine the pH and TSS of the MAT - Blue Ternate Juice Drink

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Method

This study used the quantitative experimental method or procedure involving the control or manipulation of conditions for the purpose of studying the relative effects of various treatments applied to numbers of sample, of the same treatments applied to number of different samples.

Materials and Methods

Table 1 and 2 below show the ingredients, tools, and equipment used in the study.

Table 1. Ingredients Used in the Study

Ingredients	Total Quantity
Cucumber	6kg
Sugar	5kg
Calamansi	6kg
Lemon	5pcs.
Honey	1 bottle
Blue Ternate	1kg
Agar	

Table 2. Tools and equipment used in the study

Tools	Equipment
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	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 4 of 10

Weighing Scale
 Wooden Spoon
 Strainer
 Caserole
 Glass Bottle
 Measuring cup
 Measuring spoon
 Funnel

LPG
 Sterilizer Machine
 Blender
 Refractometer
 Ph meter

Experimental Method and Design

This study was a single factor with four (4) treatment. In this study, the researcher develop and formulated their own recipe of every product for evaluation. The amount of blue ternate was manipulated in treatment A.

Experimental procedure

Sterilization of glass bottles

The bottle was prepared for packaging by washing the bottles with one drop of Zonrox bleach to sanitize. Afterwards, the bottles were placed in the sterilizer and sterilized for 10 min, removed and dried.

Preparation of Raw Materials

The lemon and calamansi were first washed under running water. Hot water was poured to the calamansi to soften its skin and was sliced. Calamansi juice was then extracted. After washing, lemon was thinly sliced and set aside.

The cucumber was washed, peeled, and diced. Afterwards, it was blended and set aside to be used for the concentrate.

For the blue ternate, the stalk of the flower was removed from the petal. The petals collected were then washed with water. The washed petals were then blended and set aside to be used for the concentrate.

The sea weed was boiled in 1L water and added with 237 mL of calamansi extract. Calamansi acted as the acid to achieve the jelly like consistency of agars and dissolve the sea weed. It was then boiled for 30 mins. Afterwards the mixture was strained to obtain the sea weed extract. The extract was set aside and cooled down prior to usage.

Preparation of concentrate

In a casserole, 1.2 L of water were poured and 250 g of lemon slices were added and brought to a boil. When boiling point was reached the lemon slices were removed. Then 1 kg sugar, 474 mL calamansi juice, 30 mL honey, and 21 mL sea weed extract was added. The mixture was boiled for another 10 mins. Next, 750 mL of the blended cucumber was added to the mixture and boiled for another 10 minutes. After 10 minutes, the mixture was removed from heat.

Procedure for Blue Ternate Juice Drink

The amount of concentrate needed (see Table 3), was measured and poured in an aluminum pan. The concentrate is heated and added with blended blue ternate petals. The volume of blue ternate added varies per treatment (see Table 3). The mixture was heated for 10 minutes, removed from heat, and poured into a sterilized glass bottle. A control sample was prepared from the concentrate without adding blue ternate.

Table 3. Treatments used for Blue Ternate Juice Drink

	T1	T2	T3	T4	T4
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	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 5 of 10

Concentrate (mL)	950	900	850	800	1000
Blue Ternate (mL)	50	100	150	200	0

Determination of Most Acceptable Treatment (MAT)

There are initially 4 treatments formulated in this study which is shown in Table 3. An acceptability test was used to determine the most acceptable treatment. This test used a 9 - point Hedonic Scale which measured the degree of likeness of for each treatment. The results were subjected to ANOVA to determine which formulation is significantly different and most acceptable.

Characterization of Blue Ternate Juice Drink

Newly developed products undergo characterization to determine its profile and understand its properties. This study determined the pH, TSS, sensorial, and microbiological characteristics of the MAT. For this study, two samples of the MAT were tested namely 1-week and 1-year storage.

A. pH

pH is a measure of a products acidity or alkalinity. In this study pH meter was used in determining the MAT's pH level.

B. TSS

TSS or total soluble solids is used to measure the sugar content of a product. In this study a refractometer was used in determining the MAT's TSS.

C. Sensory Evaluation

The MAT was subjected to sensorial characterization using a line scale (see appendix). This will determine its characteristics based on:

- Flavor
- Color
- Aroma
- Consistency

Results were tallied and mean was calculated for each characteristic. The characteristics are interpreted based on a descriptive guide provided.

D. Microbiological Test

The MAT was subjected to Aerobic Plate Count (APC) test to check its compliance with FDA safety standards. The samples were sent to DOST Region VI for testing. The laboratory followed Pour Plate Method based on USFDA BAM (Online) Standards.

Consumer Preference Test

There were two samples of MAT were subjected to consumer preference test to determine which of the two consumers and panelist prefer based on its sensorial characteristics. The panelists were given two coded samples each and were tasked to rank which of the two they prefer more.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 6 of 10

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Most Acceptable Treatment

The most acceptable treatment was determined by an acceptability test using hedonic scale. Panelists were presented with the 4 treatments and were tasked to rate each treatment based on their liking.

Results showed that treatment 4 (T4) - "800g of concentrate and 200g of blue ternate were combined in an aluminum pan and was brought to a boil for 10 minutes" was the most acceptable and showed significant difference in flavor, color, aroma, and consistency when compared to the other 3 treatments.

Physico-Chemical Tests

Table 4. Results of MAT – Blue Ternate Juice Drink pH, TSS, and APC

pH	TSS (Brix)	APC (cfu/mL)
5.13	54.33	< 1

pH

By definition, pH is the measure of a product's acidity or alkalinity. The pH level of a food product is essential in determining the survival of microorganisms. Thus, controlling the pH level of food helps in preventing its spoilage and deterioration and can prevent consumer health problems (Sper Scientific Direct, 2022).

As shown in the results the pH levels of T4 was 5.13. According to study published by Obasi et.al, fruit juices have relatively low pH levels from 1 to 5 due to the presence of organic acids. This means that the fruit juice has high acidity and high acid food products have lower risks of microorganism growth (Libretexts, 2021). Food with high acidity are easily stabilized against microbiological spoilage (Thompson, 2009).

TSS

TSS or total soluble solids indicates the level of sweetness in a food product. It is best used to measure the sugar in fruits. TSS is significantly influenced by the maturity stages and ripening conditions of the raw materials (Obasi et.al,2017).

The TSS of T4 is 54.33. The results indicate that all treatments have high TSS or sugar count. High sugar food products have lower water activity which also inhibits the growth of most spoilage and pathogenic bacteria (Thompson, 2009).

Sensory Evaluation

Sensorial characteristics of MAT was determined for profiling. A new sample with storage of 1 week and an old sample with storage of 1 year were evaluated and compared.

Table 5. Characteristics of MAT – Blue Ternate Juice Drink (1-week vs 1-year)

	Color	Taste	Odor	Consistency

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 10

1 week	5.43	3.90	4.00	4.57
1 year	3.23	3.57	4.33	4.33

Color

Color is an important factor in determining the acceptability of a product. Results showed the color has a mean of 3.90 for 1-week and 3.57 for 1-year which is equivalent to Eminence (Hex # 632673) and Vivid Violet (Hex #84319B) respectively. After 1 year of storage, a decreasing trend can be noticed for color. A decrease in color is attributed to the decrease of anthocyanins in the juice drink. As the storage extends, the number of anthocyanins decrease.

Flavor

Flavor plays a major role in a food product's acceptability. It is the combination of odor and taste sensations. In the study, panelists are tasked to rate MAT – Blue ternate juice drink from 1 to 7 and results showed a mean of 5.43 for 1-week and 3.23 for 1-year. This corresponds to a flavor of slightly sour blue ternate and slightly sweet blue ternate after taste respectively.

Blue ternate petals alone have a mildly sweet taste similar to chamomile (Slow Food, ND). Combining it with the citrus flavor of calamansi and the sweet vegetable flavor of cucumber enhanced its sweetness and acidity. A sweet-tart flavor is common in a variety of citrus mixed fruit juice.

Aroma

Aroma also aids in the acceptability of a food product. It can contribute to the perception of a food product's flavor. Results showed the aroma has a mean of 4.00 for 1-week and 4.33 for 1-year which is equivalent to moderately perceivable blue ternate aroma for both sample. The slight increase of aroma after 1 year of storage can be attributed to fermentation.

Fruits and plants have natural compounds that allow them to ferment during storage. Fermentation creates an alcoholic aroma due to the development of alcohol.

Consistency

Results showed the consistency has a mean of 4.57 for 1-week and 4.33 for 1year which is equivalent to thickened consistency for both samples. There was no significant change observed for consistency. The thickened consistency is due to the addition of sea weed which acted as a stabilizer.

Micro Tests

Presence of microorganisms in food can indicate its shelf-life, storage, and safety of consumers. Thus, monitoring of this parameter is essential in developing new food products. For this research the standard of FDA 2010-13 "Frozen Juice Concentrate" was followed. The Aerobic Plate Count (APC) was determined for the samples. An APC of 0 was detected for 1-week sample and < 1 cfu/mL was detected after a year of storage and both are within the limits of FDA.

Preference test

Table 6. Results of MAT – Blue Ternate Juice Drink Preference test (1-week vs 1-year)

	1-week	1-year
Rank Sum	13	17

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
			Revision No.	00
	Document Title:	COMPLETED RESEARCH	Effective Date	
			Page	Page 8 of 10

A preference test was conducted between the two MAT samples to determine which the panelists prefer the most. Results indicated that MAT with 1-year storage was most preferred by the panelists.

Fruit juices and concentrates generally ferments during storage. Fermentation further develops the sensorial characteristics of the juice which have contributed to the preference of panelists for the 1-year sample.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

After conducting the study, results show that the most acceptable treatment is T4 based on sensory characteristics of flavor, color, aroma, and consistency. Characterization of the MAT was conducted and results showed it has an eminence color, slightly sour flavor, moderate perceivable aroma, and thickened consistency. A sample of MAT was stored for 1 year and was also characterized. Results showed it has a vivid violet color and slightly sweet after taste for flavor while its aroma and consistency remained the same. A preference test was conducted for the two MATs and panelists preferred the 1-week sample.

Physico-chemical tests such as pH and TSS were also conducted for 1-week sample. Results show a pH of 5.13 and TSS of 54.33 which is within acceptable limits for fruit juice concentrates.

A microbiological test for APC was also conducted. It showed an APC count of 0 for 1-week sample and <1 cfu/ml after 1 year of storage. This indicates that after 1 year of storage, it is safe to consume and within the standards set by FDA.

Therefore, Blue Ternate (*Clitoria ternate*) is acceptable in the development of an alternative juice drink. To further improve the product and for future researchers who would like to continue this study the following recommendations are made:

1. Test for the yeast and molds count of the MAT to enhance its shelf life;
2. Test for the acidity of the MAT through Titratable Acidity for profiling of physico-chemical characteristics; and
3. Determine the feasibility of sea weed as a natural stabilizer
4. Test the feasibility of blue ternate as natural colorant

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

Blue Ternate Concentrate

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

Baclig, C. 2022. " World Food Day 2022: Rising costs keep millions in PH away from healthy diets". Retrieved from:

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 9 of 10

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	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 10 of 10

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